
State of the Art

Crowd Sentiment Analysis for Stock Market Prediction and Market Manipulation Detection

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Abstract

This state of the art surveys the research literature most relevant to the CrowdSignal project, a real-time platform that aggregates crowd sentiment from social media platforms, scores it with the FinBERT financial NLP model, and cross-references it against live market data to detect potential market manipulation. We review four interconnected threads: the theoretical foundations of crowd wisdom in financial markets, social media as a predictive signal source, NLP models for financial sentiment analysis with emphasis on FinBERT-based approaches, and the detection of social media-driven market manipulation using the GameStop 2021 event as the primary case study.

1 Introduction

Financial markets are information-processing systems: asset prices reflect the collective interpretation of available information by all market participants. Traditionally, this information came from fundamentals — earnings reports, macroeconomic indicators, and analyst forecasts. The rise of the internet, and social media in particular, introduced a new class of signal: the aggregated, real-time opinion of millions of retail investors expressing views in plain language online.

The intersection of natural language processing, social media analytics, and financial markets has grown into one of the most prolific research areas in computational finance over the past two decades. The foundational hypothesis is that the aggregated opinions of a large online crowd carry predictive signal about future asset price movements — a

computational instantiation of the *wisdom of crowds* concept popularized by ?. Landmark empirical work by ? demonstrated that Twitter-derived public mood Granger-causes movements in the Dow Jones Industrial Average with an 86.7% directional accuracy, establishing social media sentiment as a legitimate financial signal.

More recent research has expanded beyond Twitter to dedicated financial platforms such as StockTwits [Liu et al.] and online investor forums [Dong et al.], explored advanced NLP architectures [?Ruan and Jiang], and investigated the darker side of social aggregation: coordinated market manipulation through crowd action [La Morgia et al., ?].

This review is structured as follows. Section 2 covers the theoretical foundations of crowd wisdom and its aggregation methodologies. Section 3 surveys social media as a market signal source. Section 4 examines NLP models for financial sentiment, with focus on FinBERT. Section 5 analyzes social media-driven market manipulation. Section ?? positions CrowdSignal relative to existing work, and Section ?? derives testable research hypotheses.

2 Theoretical Foundation: Wisdom of the Crowd in Financial Markets

2.1 From Individual Judgment to Collective Intelligence

Before examining how crowds behave in financial markets, it is useful to understand the broader concept of *collective intelligence* — the phenomenon by which a group of individuals, each with partial and imperfect information, can together produce estimates or decisions that outperform those of any single expert. This is not a financial phenomenon; it emerges in biology, economics, and organizational theory alike. The key insight is that *diversity of independent signals*, when properly aggregated, cancels out individual errors and surfaces the true underlying value.

Applied to financial markets, this principle is formalized in ?’s *Wisdom of the Crowds*. The theory posits that diverse retail investors, each holding a fragment of private information, produce an aggregate signal more accurate than any individual forecast, provided three conditions hold: *diversity* (participants hold genuinely different views), *independence* (opinions are not imitated from others), and *decentralization* (no single entity controls the group). Critically, the theory also identifies the failure mode: crowds become irrational when independence breaks down through herding or coordinated action — precisely the failure mode that CrowdSignal’s manipulation detection layer targets.

2.2 Crowd Aggregation in Practice

Dong et al. extend this framework by proposing a novel *artificial prediction market* constructed from comment posters on the Eastmoney stock forum. Rather than using traditional poll-based aggregation, which is vulnerable to low-quality inputs, their mechanism uses a market-pricing dynamic to filter unreliable signals. Validated on the Shanghai Stock Exchange 50 Index constituents, the system demonstrates that well-designed crowd aggregation significantly outperforms prediction polls in extracting signal from noisy online communities.

The aggregation step itself is the subject of a dedicated systematic review by Liu and Son. Examining mainstream methods for constructing investor sentiment indices from social media, they conclude that the choice of aggregation algorithm directly determines the reliability of the resulting signal. They identify a critical and frequently overlooked issue: the treatment of *neutral* sentiment texts. Naïve aggregation that ignores neutral signals or misclassifies them introduces systematic noise. This finding has direct methodological implications for CrowdSignal, which uses a sliding 30-minute window average of FinBERT scores and must handle the proportion of neutral-labeled posts carefully to avoid biasing the aggregate sentiment signal.

3 Social Media as a Stock Market Predictor

3.1 Information Sources and Market Signals

In financial markets, a *market signal* is any observable variable that carries predictive information about future price movements. Classical signals include order flow, trading volume, and earnings surprises. The question motivating this section is whether the text produced by retail investors on public platforms constitutes a reliable signal of this kind.

The theoretical basis is straightforward: if a large number of investors publicly express their intentions or beliefs before acting on them, the aggregate of those expressions should anticipate the aggregate of their trades. Social media platforms — first general ones such as Twitter, then dedicated financial communities such as StockTwits and Reddit r/WallStreetBets — have become the primary venues for this public expression. The research surveyed below traces the empirical validation of this idea, from early Twitter studies to platform-specific deep learning pipelines.

3.2 From Twitter to Dedicated Investor Platforms

The foundational empirical study in this field is ?, who analyzed daily Twitter feeds using OpinionFinder and the Google-Profile of Mood States (GPOMS). Their Granger causality analysis showed that the “Calm” mood dimension preceded DJIA changes by three days

with 86.7% directional accuracy, establishing that public mood derived from general social media carries genuine market-predictive information.

Affuso and Lahtinen, in *Empirical Economics*, examined the geographic and asymmetric dimensions of Twitter sentiment on individual stock returns. Their key finding is that negative tweets exert a significantly larger effect on returns than positive ones (0.078% vs. 0.036% daily impact, respectively). This asymmetry is directly relevant to CrowdSignal’s signal design: bearish signals derived from the consistency engine should be acted upon with higher confidence than bullish ones, given that negative crowd sentiment has stronger predictive force.

Saravanos and Kanavos, in *Neural Computing and Applications*, present a deep neural network model integrating historical market data with Twitter sentiment and an extensive array of tweet metadata features. Contrary to concerns about high-dimensional feature sparsity, they find that incorporating features such as tweet diffusion and the influence of prominent accounts consistently improves forecasting accuracy. This validates the multi-feature aggregation approach used in CrowdSignal, where post volume, source platform, and recency all contribute to the final aggregate signal.

Kim-Hahm et al., in the *Journal of Risk and Financial Management*, compare the sentiment impact of traditional news media, Twitter, and web search on the stock performance of Apple, Amazon, and Microsoft over 2015–2020. Applying FinBERT-based analysis across all three sources, they find that social media sentiment exerts a consistently positive and significant influence on stock returns, while traditional news effects are more irregular. This finding strongly supports the choice of social platforms, rather than financial news feeds, as the primary signal source for CrowdSignal.

3.3 StockTwits as a Dedicated Financial Signal

Liu et al., in *PeerJ Computer Science*, specifically investigate StockTwits as a signal source for predicting SPDR S&P 500 ETF movements. Their pipeline uses FinBERT for sentiment scoring and an ensemble SVM classifier with a rolling window approach. FinBERT achieves an F1-score 4–5% higher than competing sentiment methods on StockTwits data. A particularly valuable property of StockTwits highlighted in this study is that users voluntarily self-tag a subset of messages as bullish or bearish, providing a natural validation ground truth. CrowdSignal explicitly exploits this property: user-tagged StockTwits labels are stored alongside FinBERT predictions to measure model agreement and serve as a cross-validation signal.

3.4 Forum-Based Sentiment with Deep Learning

Mu et al., in *IEEE Access*, integrate a custom sentiment dictionary derived from Eastmoney forum posts with technical trading data into a Sparrow Search Algorithm-optimized LSTM model (MS-SSA-LSTM). Their results show that adding a sentiment index improves prediction R^2 by 10.74% on average compared to standard LSTM, and that the combination of sentiment with fundamental trading data outperforms either signal alone. This reinforces the cross-referencing design of CrowdSignal: sentiment alone is not sufficient; the signal gains predictive value when combined with market ground truth.

4 NLP Models for Financial Sentiment Analysis

4.1 From Text to Structured Sentiment: A Conceptual Path

Natural Language Processing (NLP) is the branch of artificial intelligence concerned with enabling machines to interpret and analyze human language. Within NLP, *sentiment analysis* is the task of automatically determining whether a piece of text expresses a positive, negative, or neutral opinion. Applied to finance, this general capability must be adapted to handle domain-specific vocabulary, since words that are neutral in everyday language (“liability,” “downgrade,” “miss”) carry strong directional meaning in financial contexts.

The path from raw text to a usable financial sentiment score has evolved through three generations of methods. The first generation relies on domain-specific *lexicons* (e.g., the Loughran-McDonald dictionary), which assign polarity based on word lists. The second generation applies *machine-learning classifiers* (Naïve Bayes, SVM) trained on labeled financial text, achieving higher accuracy at the cost of requiring curated datasets. The third and current generation leverages *pre-trained transformer models* through transfer learning, dramatically reducing the labeled data requirement while achieving state-of-the-art performance.

4.2 FinBERT: The Domain-Adapted Standard

The model central to CrowdSignal’s pipeline is FinBERT, introduced by ? in *FinBERT: Financial Sentiment Analysis with Pre-trained Language Models* (arXiv:1908.10063). By further pre-training BERT on the Reuters TRC2 financial news corpus and fine-tuning on two labeled financial sentiment datasets, FinBERT achieves a 14-percentage-point improvement over prior state-of-the-art methods. The key advantage is the model’s internalization of financial jargon — terms such as “beat,” “miss,” “guidance,” and “downgrade” carry directional sentiment that general-purpose models systematically misinterpret. The `yyanghkust/finbert-tone` variant available on HuggingFace produces three output

probabilities (Positive, Negative, Neutral), and CrowdSignal derives a continuous score as $\text{score} = P(\text{positive}) - P(\text{negative}) \in [-1.0, +1.0]$.

4.3 Recent FinBERT-Based Frameworks

Ruan and Jiang, in *Mathematics*, propose a multimodal prediction framework combining FinBERT-derived sentiment from financial news headlines with technical and statistical indicators, fed into an XGBoost classifier. Their SHAP analysis reveals that FinBERT-derived features consistently rank among the top predictors across multiple S&P 500 equities over 2018–2023, outperforming both technical-only and lexicon-based baselines in AUC, F1-score, and simulated trading profitability. This provides direct empirical validation that FinBERT sentiment adds predictive power above and beyond price-based signals — the central design assumption of CrowdSignal.

Kim-Hahm et al. apply FinBERT across three media source types and find consistent sentiment-return links for social media channels, further validating the applicability of FinBERT to the type of short-form financial text found on Twitter and StockTwits.

4.4 Limitations and the Slang Problem

A key limitation identified across multiple studies is that FinBERT, trained primarily on structured financial news text, underperforms on informal social media language. Reddit posts use heavy slang, emojis, irony, and meme-culture vocabulary (“tendies,” “moon,” “diamond hands”) outside FinBERT’s training distribution. This is precisely why CrowdSignal applies a slang normalization step before scoring, mapping Reddit-specific vocabulary to plain financial English (e.g., “moon” → “very bullish increase”) before passing text to the model. Liu and Son similarly recommend that preprocessing choices be explicitly documented and justified, noting that undisclosed preprocessing differences are a major source of inconsistency across the literature.

5 Social Media-Driven Market Manipulation

5.1 Market Manipulation: From Definition to Detection

Market manipulation refers to any deliberate attempt to interfere with the free and fair operation of financial markets, typically by artificially inflating or deflating the price of an asset to profit at the expense of other participants. Regulatory frameworks (SEC, MiFID II) classify manipulation into several forms: *spoofing* (placing and canceling large orders to move prices), *wash trading* (buying and selling the same asset to create artificial volume), and *pump and dump* (coordinated promotion of an asset to inflate its price before selling).

The emergence of social media introduced a new vector: *coordinated crowd manipulation*,

where a large number of participants are recruited through public online channels to execute trades simultaneously, creating price and volume anomalies that mimic organic market activity and are therefore harder to detect with classical surveillance tools. The literature surveyed below narrows progressively from manipulation in general, to social media-driven schemes, to the specific pump-and-dump pattern, and finally to the landmark GameStop case that defines CrowdSignal’s primary detection scenario.

5.2 Pump and Dump in Crypto Markets

La Morgia et al., in *ACM Transactions on Internet Technology*, provide the most rigorous empirical study of pump and dump schemes on cryptocurrency markets. Monitoring Telegram and Discord groups for over three years, they compiled a dataset of approximately 900 individually verified pump and dump events. Key findings include:

- Pump and dump follows a consistent temporal signature: sharp volume spike, rapid price increase, followed by an equally rapid reversal.
- The authors build a real-time machine learning detector achieving a 94.5% F1-score, capable of flagging an event within **25 seconds** of its initiation.
- They also characterize the *crowd pump* — a newer phenomenon where a Reddit community inflated GameStop stock by over 1,900% and later replicated the operation on DogeCoin and Ripple.

The feature engineering used in their detector — volume anomaly, sentiment spike, and price momentum divergence — maps directly to CrowdSignal’s consistency score: when crowd sentiment diverges sharply from price momentum, the system fires an `AlertFlag`. Their work validates this detection approach on real manipulation events, providing a strong empirical precedent.

5.3 The GameStop Case Study

The GameStop (GME) short squeeze of January 2021 is the canonical case of social media-coordinated market action. Members of `r/wallstreetbets` coordinated purchases of GME shares, triggering a short squeeze that drove the price from approximately \$7 to a peak of \$347.51 — a 1,700% increase in days, costing hedge fund Melvin Capital nearly \$3 billion.

?, in *Financial Review*, conducted the most rigorous academic analysis, processing 10.8 million `r/WallStreetBets` comments at 1-minute resolution with a Reddit-specific VADER lexicon. Their Granger causality analysis at [1-, 5-, 10-, and 30-minute] frequencies found that:

- Post *volume* was a significant leading indicator of GME price movements at short

intraday frequencies.

- Average sentiment *alone* did not Granger-cause returns at any tested frequency.
- The *interaction* between volume extremity and sentiment intensity was the most informative signal.

This finding directly informs CrowdSignal’s design: the system tracks both sentiment score and post count within each aggregation window, and the alert threshold requires a minimum of 5 posts to avoid triggering on isolated noise.

?, in *Scientific Reports*, modeled the self-reinforcing dynamics of Reddit consensus formation. They showed that GME-related comment frequency and sentiment increased significantly *before* the short squeeze — i.e., the social signal was an early warning, not merely a concurrent indicator. Their opinion dynamics model reveals a *phase transition* from heterogeneous to homogeneous crowd opinions as engagement grows, which they link qualitatively to the sudden price hike. This is the theoretical justification for CrowdSignal’s manipulation detection layer: a rising consistency between social platforms (Reddit amplifying StockTwits) combined with divergence from price momentum is an early warning of emerging coordinated action.

6 Summary of Reviewed Literature

Table 1 summarizes each paper reviewed in this state of the art and its specific relevance to the CrowdSignal project.

Paper	Relation to CrowdSignal
Surowiecki (2004)	Theoretical foundation: crowd wisdom conditions (diversity, independence, decentralization) and failure modes that CrowdSignal’s manipulation layer targets.
Bollen et al. (2011)	Established Twitter mood as a market predictor (86.7% accuracy on DJIA), validating social sentiment as a financial signal.
Affuso & Lahtinen (2019)	Showed negative tweets have stronger price impact than positive ones, informing asymmetric confidence weighting in CrowdSignal.
Araci (2019)	Introduced FinBERT, the core NLP model in CrowdSignal’s sentiment scoring pipeline.
Long et al. (2022)	Reddit WSB analysis at 1-min resolution during GameStop; showed volume×sentiment interaction is the key signal, directly shaping CrowdSignal’s dual-metric alert design.
Campajola et al. (2022)	Modeled Reddit opinion phase transitions before the GME squeeze, providing theoretical basis for CrowdSignal’s cross-platform consistency-based manipulation detection.
La Morgia et al. (2023)	Pump-and-dump detection on crypto (94.5% F1 in 25s); feature engineering (volume anomaly, sentiment spike, momentum divergence) maps directly to CrowdSignal’s alert logic.
Liu et al. (2023)	FinBERT on StockTwits with ensemble SVM; validated StockTwits self-tags as ground truth, which CrowdSignal uses for cross-validation.
Mu et al. (2023)	Forum sentiment + LSTM improved prediction R^2 by 10.74%; confirms sentiment gains value when combined with market data, supporting CrowdSignal’s cross-referencing design.
Dong et al. (2024)	Artificial prediction market for crowd aggregation on forums; demonstrates market-based filtering outperforms polls for noisy communities.
Liu & Son (2024)	Systematic review of sentiment aggregation methods; identified neutral-text handling as critical, directly relevant to CrowdSignal’s sliding-window averaging.
Kim-Hahm et al. (2025)	Compared news vs. social media with FinBERT; social media sentiment more consistently predictive, supporting CrowdSignal’s platform choice.
Ruan & Jiang (2025)	FinBERT + XGBoost with SHAP; FinBERT features rank top predictors across S&P 500, empirically validating CrowdSignal’s core design assumption.
Saravanos & Kanavos (2025)	Twitter metadata + DNN for market forecasting; validates multi-feature aggregation (volume, source, recency) used in CrowdSignal.

Table 1: Summary of reviewed papers and their relation to the CrowdSignal project.

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